OpenShift Data Foundation (ODF) Deployment

1. Overview

This document provides a production-grade procedure for deploying OpenShift Data Foundation (ODF) on a connected OpenShift Container Platform (OCP) cluster. It includes prerequisites, installation steps, validation, and troubleshooting guidance.

2. Architecture Overview

ODF provides the following storage services:

- Block Storage (RBD)
- File Storage (CephFS)
- Object Storage (NooBaa S3 compatible)

Core components:

- Ceph MON (Monitor)
- Ceph MGR (Manager)
- Ceph OSD (Storage Daemons)
- Ceph MDS (File system metadata)
- NooBaa (Object storage layer)

3. Prerequisites

3.1 OpenShift Cluster Requirements

- OCP version: 4.12 or later (recommended 4.14+)
- Minimum 3 worker nodes (for production HA)
- Cluster must be in healthy state

Validation:

```
oc get nodes
```

```
oc get clusterversion
```

3.2 Hardware Requirements

Each worker node must have:

- Minimum 16–32 GB RAM
- Minimum 4 CPUs
- Raw unformatted disks (SSD recommended)

Example disks:

- /dev/sdb
- /dev/sdc
- /dev/sdd

3.3 Network Requirements

- Full internal connectivity between nodes
- API and ingress reachable
- DNS resolution working for cluster domain

3.4 Operators Required

- OpenShift Data Foundation Operator
- Local Storage Operator

4. Pre-Checks

4.1 Node health check

```
oc get nodes -o wide
```

4.2 Cluster operators health

```
oc get co
```

5. Installing ODF Operator

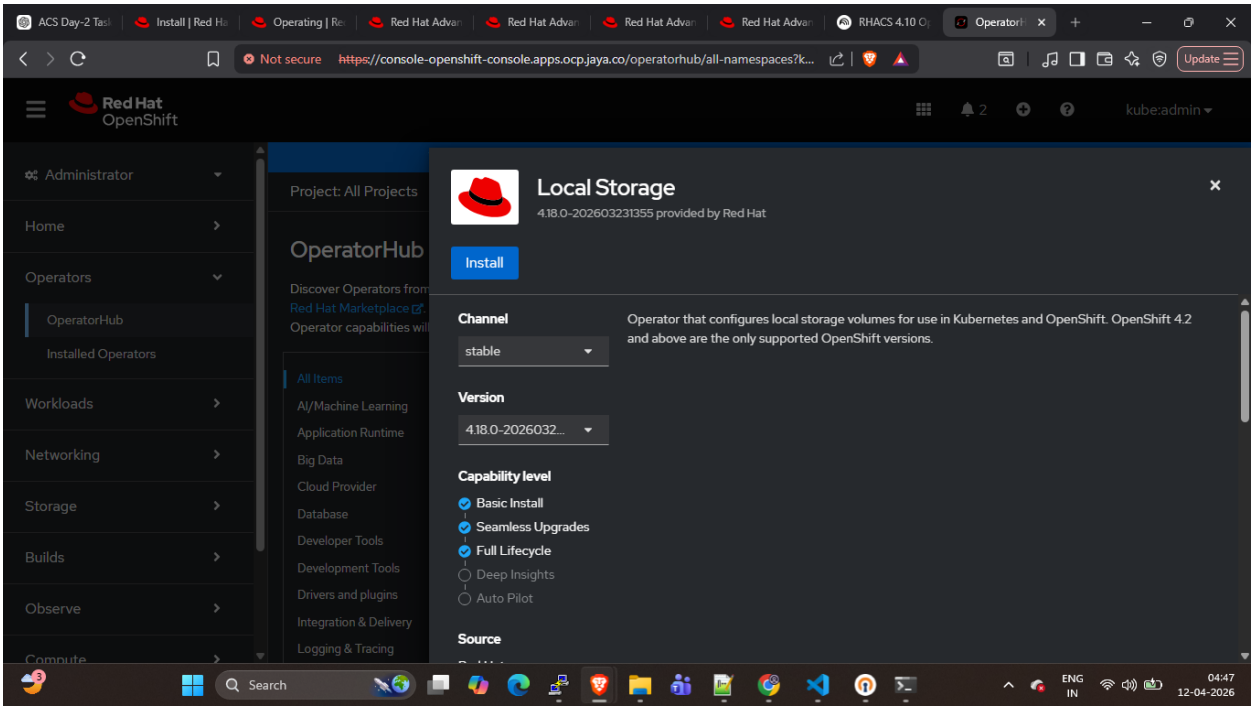
Step 1: Open OperatorHub

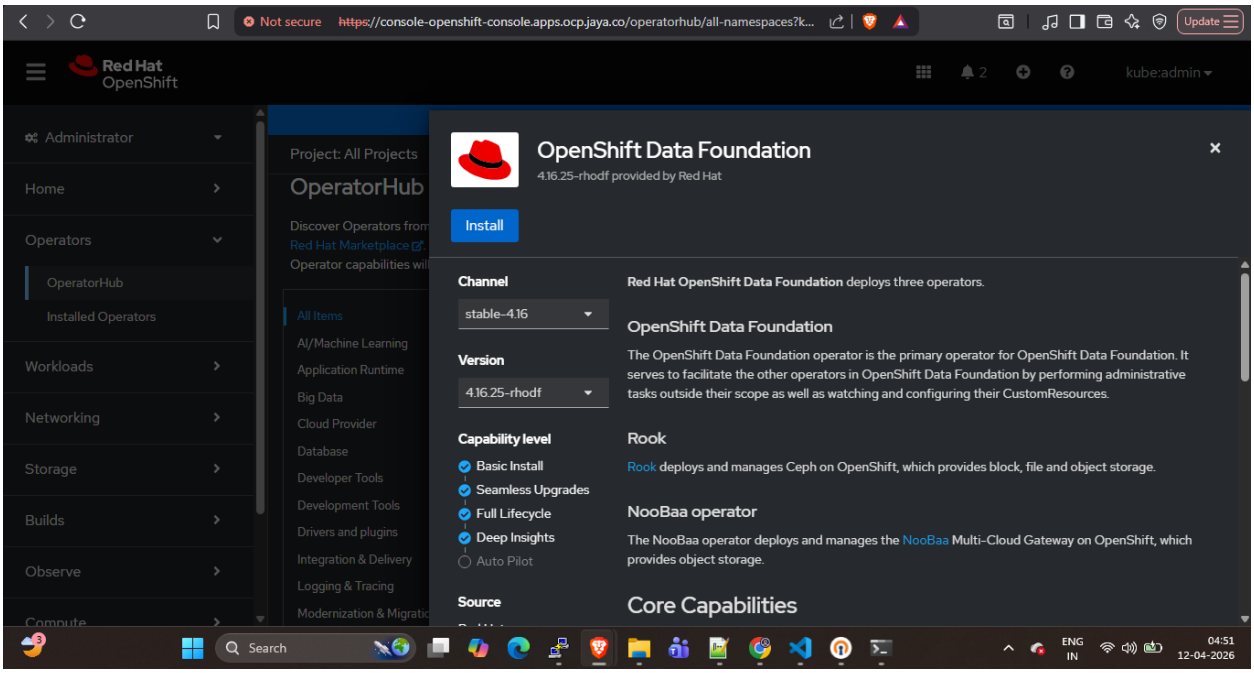
Go to:
OpenShift Console → Operators → OperatorHub

Step 2: Search Local storage and ODF

Search for:

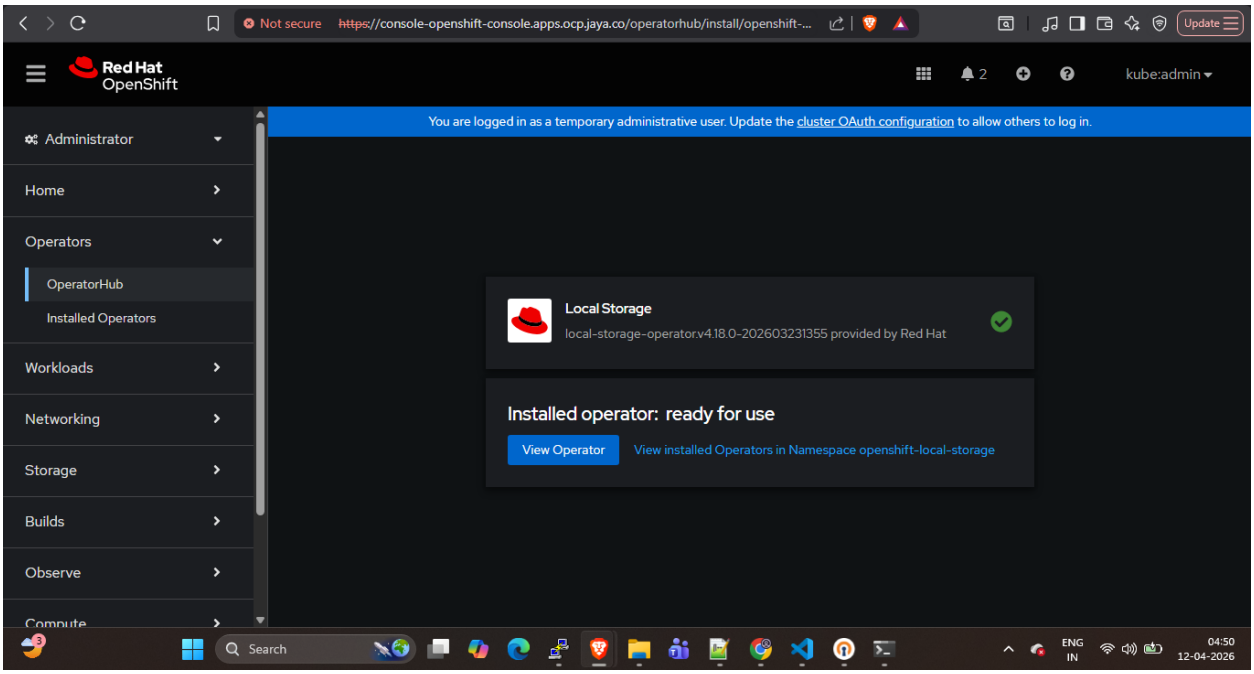
Local storage
OpenShift Data Foundation

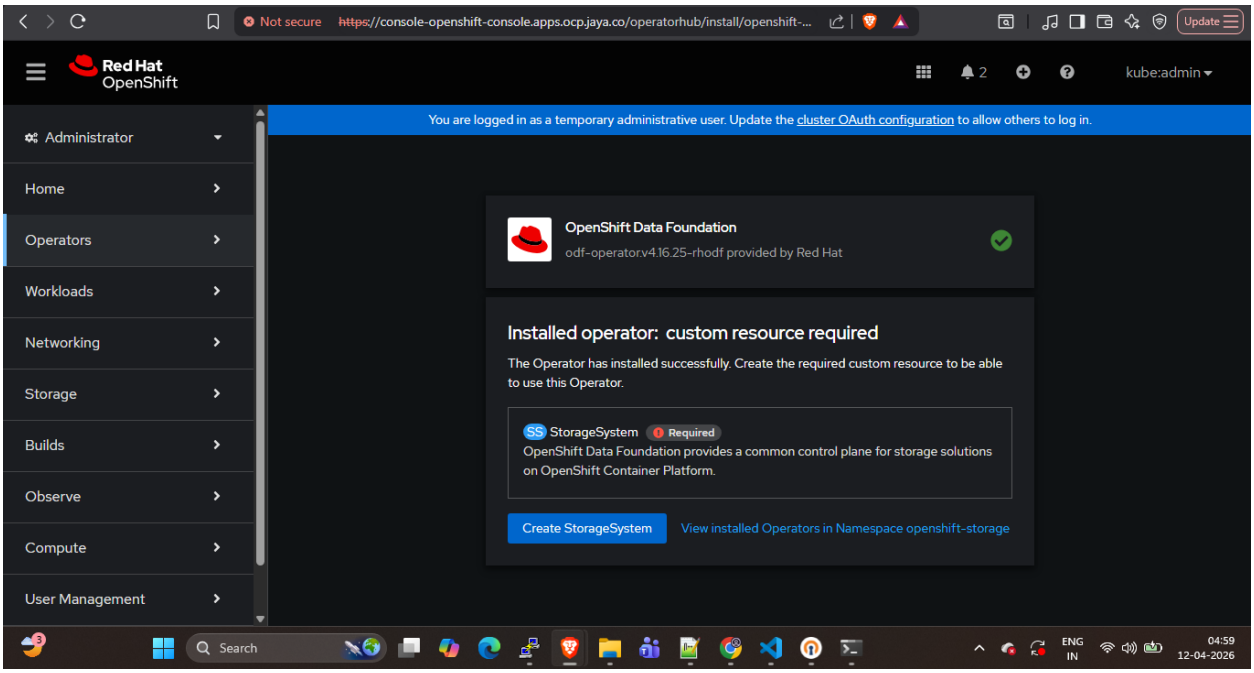




Step 3: Install Operator

- Update channel: stable
- Installation mode: All namespaces
- Approval: Automatic





I verified by following the below methods

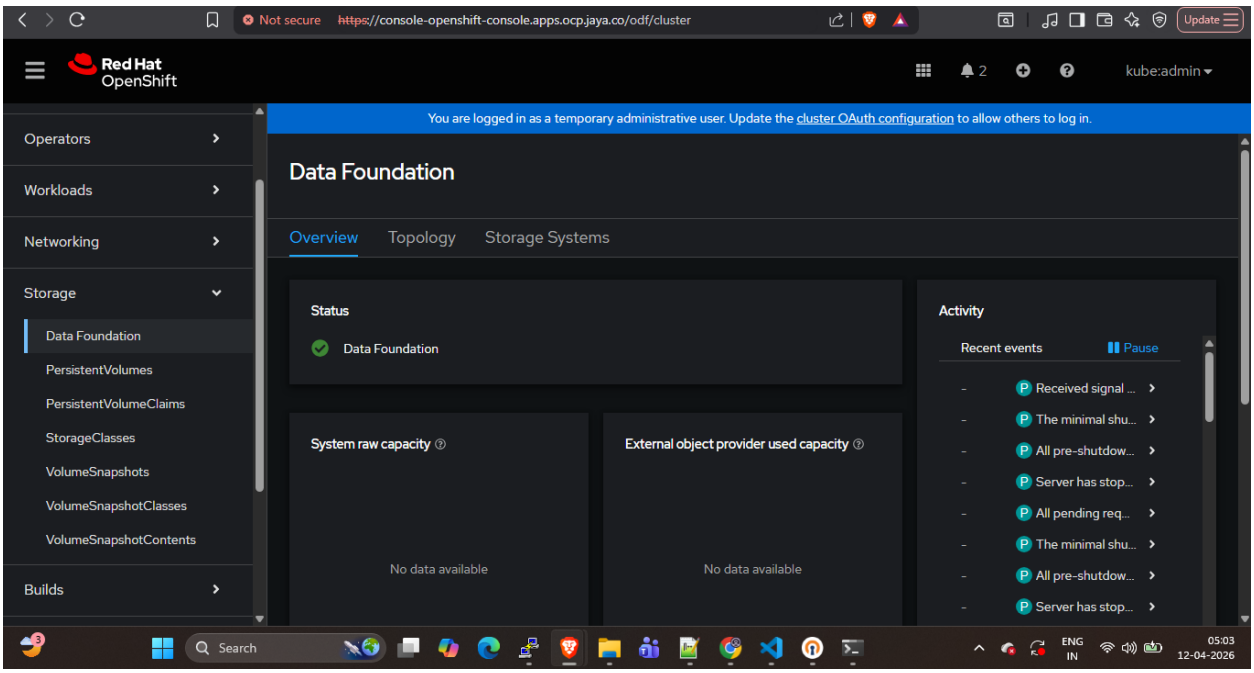
```
[root@bastion install]# oc get pods -n openshift-storage
NAME                                          READY   STATUS    RESTARTS        AGE
csi-addons-controller-manager-5b87cfd955-x8s4f   2/2     Running   1 (2m47s ago)   3m50s
noobaa-operator-85cf956b5b-647zj              1/1     Running   0               4m45s
ocs-operator-848854984d-jvdtn                 1/1     Running   1 (2m48s ago)   4m28s
odf-console-574cdf8795-kjb7t                  1/1     Running   0               4m22s
odf-operator-controller-manager-6b7bb564-6jq9h   2/2     Running   1 (2m47s ago)   4m22s
rook-ceph-operator-954bcb77c-qt4wh            1/1     Running   0               4m3s
ux-backend-server-5fcfb65bb7-d7jvm            2/2     Running   0               4m28s
[root@bastion install]#
```

```
[root@bastion install]# oc get deployment -n openshift-storage
NAME                                   READY   UP-TO-DATE   AVAILABLE   AGE
csi-addons-controller-manager          1/1     1            1           5m9s
noobaa-operator                        1/1     1            1           5m11s
ocs-client-operator-console            0/0     0            0           4m57s
ocs-client-operator-controller-manager 0/0     0            0           4m57s
ocs-operator                           1/1     1            1           4m54s
odf-console                            1/1     1            1           4m48s
odf-operator-controller-manager        1/1     1            1           4m48s
prometheus-operator                    0/0     0            0           4m51s
rook-ceph-operator                     1/1     1            1           4m30s
ux-backend-server                      1/1     1            1           4m54s
[root@bastion install]#
```

6. Creating Storage System

Step 1: Navigate to Storage

OpenShift Console → Storage → Data Foundation



Step 2: Create StorageSystem

Click:
Create StorageSystem

Select:

- Full deployment

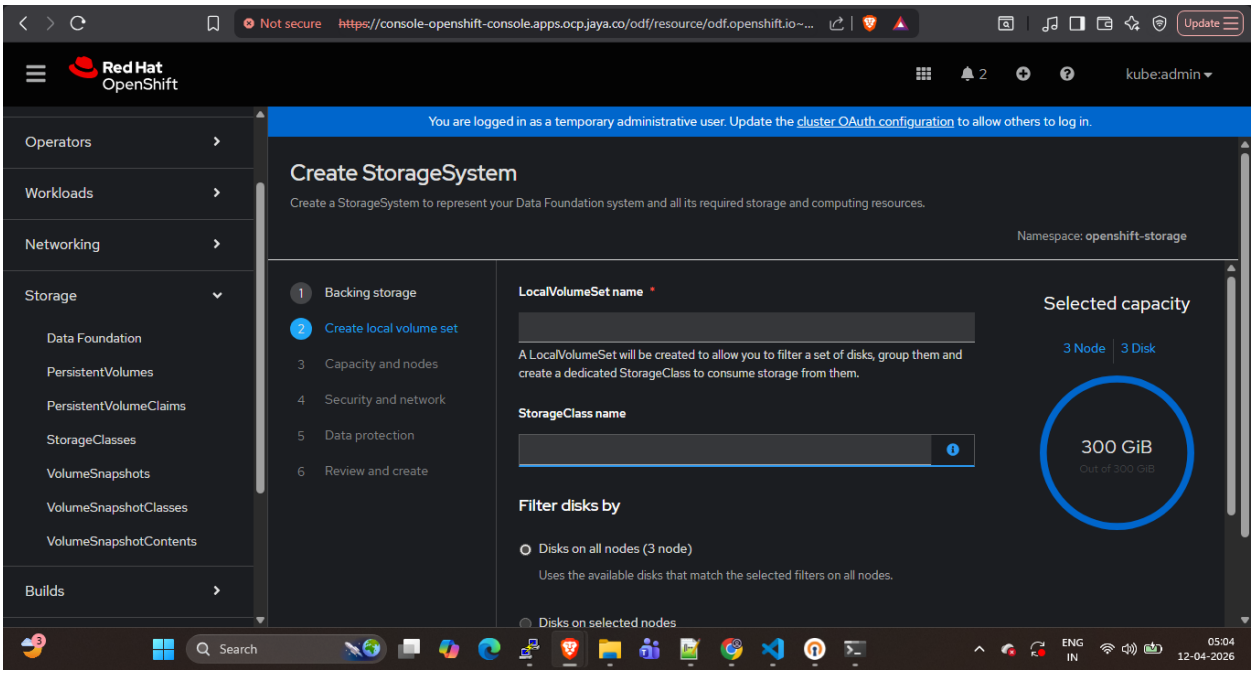
7. Node and Disk Selection

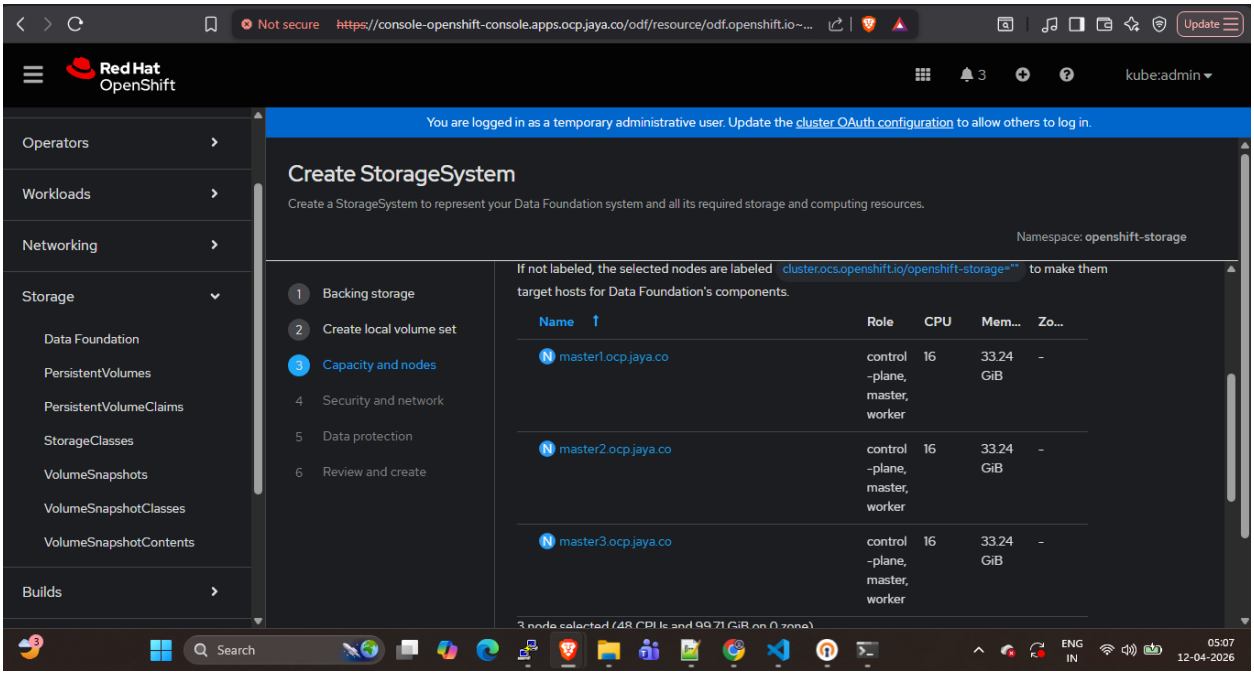
Step 1: Select Nodes

Click on create storage system

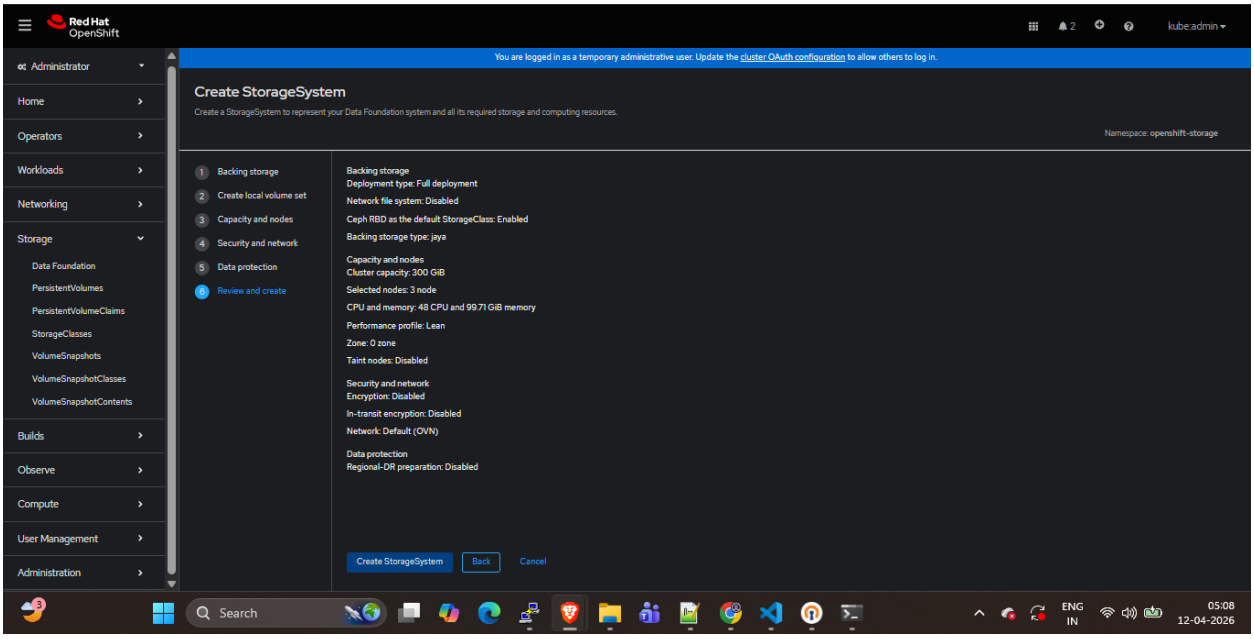
Step 2: Give name for LV

Step 3: ODF will detect new disks

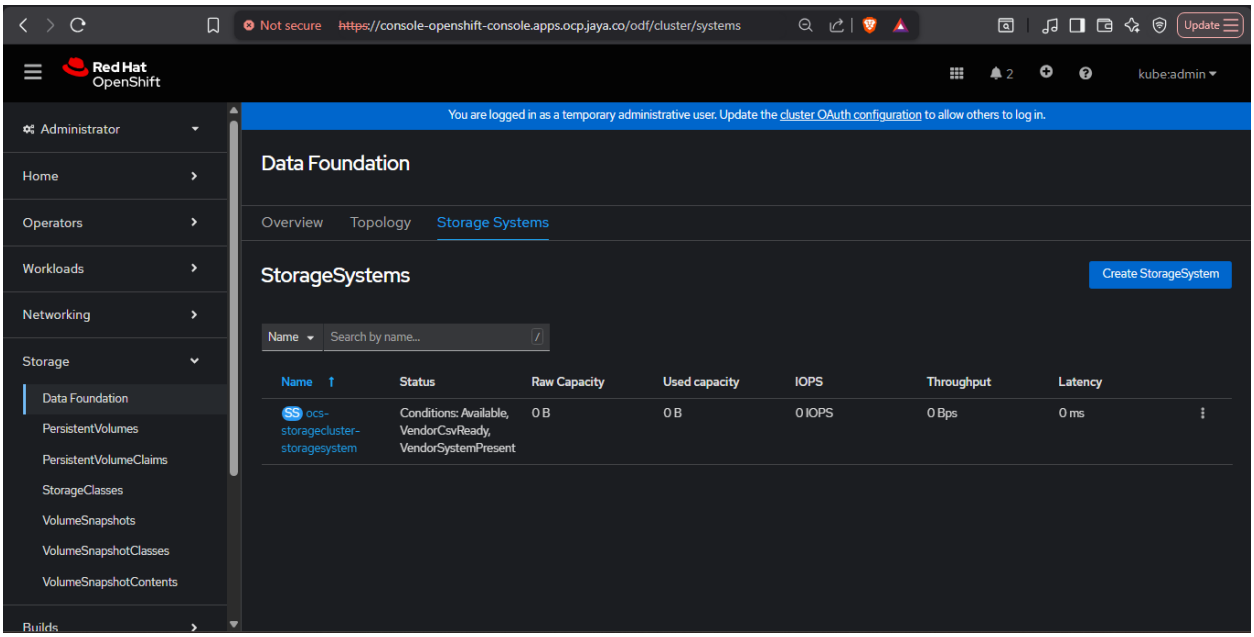




Step 4: follow through the process and review and final



8. Validation Steps



8.1 Check Pods

```
oc get pods -n openshift-storage
[root@bastion install]# oc get pods -n openshift-storage
NAME                                                                    READY
STATUS       RESTARTS         AGE
```

```
csi-addons-controller-manager-5b87cfd955-x8s4f                    2/2
Running     2 (4m17s ago)   20m
csi-cephfsplugin-4bgtv                                            2/2
Running     0               5m27s
csi-cephfsplugin-97qdm                                            2/2
Running     0               5m27s
csi-cephfsplugin-c2994                                            2/2
Running     1 (4m42s ago)   5m27s
csi-cephfsplugin-provisioner-6b88d6db64-4gr2t                     6/6
Running     0               5m27s
csi-cephfsplugin-provisioner-6b88d6db64-vl6cb                     6/6
Running     2 (80s ago)     5m27s
csi-rbdplugin-4pb8k                                               3/3
Running     2 (66s ago)     5m27s
csi-rbdplugin-4wgsd                                               3/3
Running     0               5m27s
csi-rbdplugin-5xmdk                                               3/3
Running     1 (96s ago)     5m27s
csi-rbdplugin-provisioner-864ccb678c-4954z                        6/6
Running     4 (63s ago)     5m27s
csi-rbdplugin-provisioner-864ccb678c-8rdbg                        6/6
Running     2 (96s ago)     5m27s
noobaa-core-0                                                     2/2
Running     0               45s
noobaa-db-pg-0                                                    1/1
Running     0               45s
noobaa-operator-85cf956b5b-647zj                                  1/1
Running     0               21m
ocs-metrics-exporter-6566b87f65-h4w4j                             1/1
Running     0               40s
ocs-operator-848854984d-jvdtn                                     1/1
Running     3 (4m3s ago)    20m
odf-console-574cdf8795-kjb7t                                      1/1
Running     0               20m
odf-operator-controller-manager-6b7bb564-6jq9h                    2/2
Running     3 (4m24s ago)   20m
rook-ceph-crashcollector-master1.ocp.jaya.co-764d879556-cwdth     1/1
Running     0               2m6s
rook-ceph-crashcollector-master2.ocp.jaya.co-5cd89d7bf-88mwv      1/1
Running     0               68s
rook-ceph-crashcollector-master3.ocp.jaya.co-7747d57b-pxhms       1/1
Running     0               67s
rook-ceph-exporter-master1.ocp.jaya.co-f7d5f7949-xqrw2            1/1
```

```
Running     0               2m3s
rook-ceph-exporter-master2.ocp.jaya.co-54f9486fbd-fg9n2           1/1
Running     0               65s
rook-ceph-exporter-master3.ocp.jaya.co-75bc867cf7-l5nzf           1/1
Running     0               63s
rook-ceph-mds-ocs-storagecluster-cephfilesystem-a-f5d8cbdfgt5pl   2/2
Running     0               68s
rook-ceph-mds-ocs-storagecluster-cephfilesystem-b-548b95592s68s   2/2
Running     0               67s
rook-ceph-mgr-a-56ff4c8d9d-4phs7                                  3/3
Running     0               2m43s
rook-ceph-mgr-b-67c696f7dc-dl2td                                  3/3
Running     0               2m42s
rook-ceph-mon-a-865969b7b5-jf648                                  2/2
Running     0               3m33s
rook-ceph-mon-b-c767796cc-6fcq2                                   2/2
Running     0               3m5s
rook-ceph-mon-c-6565d58984-dhxjs                                  2/2
Running     0               2m54s
rook-ceph-operator-954bcb77c-qt4wh                                1/1
Running     0               20m
rook-ceph-osd-0-7c7ffb994c-k6r6h                                  2/2
Running     0               2m8s
rook-ceph-osd-1-ff57bd48f-2swsn                                   2/2
Running     0               2m7s
rook-ceph-osd-2-84945478b9-xmgsf                                  2/2
Running     0               2m7s
rook-ceph-osd-prepare-232cbabe63fb3107ab26caa9c37fea62-z4djl      0/1
Completed   0               2m20s
rook-ceph-osd-prepare-5153d2ee472cb0a2276ab76b75d839d7-ckz9c      0/1
Completed   0               2m20s
rook-ceph-osd-prepare-626003c651246775544c94f8a04837ce-qdfct      0/1
Completed   0               2m20s
rook-ceph-rgw-ocs-storagecluster-cephobjectstore-a-86ddc4frsnhj   2/2
Running     0               47s
ux-backend-server-5fcfb65bb7-d7jvm                                2/2
Running     0               20m
[root@bastion install]#
```

8.2 Check StorageCluster

```
oc get storagecluster -n openshift-storage
[root@bastion install]# oc get storagecluster -n openshift-storage
NAME                 AGE   PHASE         EXTERNAL   CREATED AT
VERSION
ocs-storagecluster   6m    Progressing              2026-04-11T23:38:43Z
4.16.25
```

Expected:

- Phase: Ready

8.3 Check StorageClasses

```
oc get sc
[root@bastion install]# oc get sc
NAME                                   PROVISIONER
RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION   AGE
jaya                                   kubernetes.io/no-provisioner
Delete          WaitForFirstConsumer   false                  9m14s
ocs-storagecluster-ceph-rbd (default)   openshift-storage.rbd.csi.ceph.com
Delete          Immediate              true                   2m5s
ocs-storagecluster-ceph-rgw             openshift-
storage.ceph.rook.io/bucket   Delete          Immediate              false
6m32s
ocs-storagecluster-cephfs               openshift-
storage.cephfs.csi.ceph.com   Delete          Immediate              true
109s
openshift-storage.noobaa.io             openshift-storage.noobaa.io/obc
Delete          Immediate              false                  30s
[root@bastion install]#
```